Autonomous Agent Architecture & Governance Spec

Standard Operating Guidelines for Multi-Model RAG & Execution Engines

1. Document Governance & Authorship

Document ID: APX-SEC-2026-991
Revision: 4.2.0-RC1 (Effective: October 2026)
Lead Architect: Dr. Marcus Vance (Director of Systems Architecture)
Primary Contact Email: m.vance@apexcore-security.internal
Emergency Escalation Phone: +1-555-019-2834
Security Officer: Elena Rostova (VP of Cloud Risk Governance)
Classification: Highly Confidential / Enterprise Internal Tier-1

2. System Overview & Core Capabilities

The ApexCore Autonomous System operates a decoupled retrieval-augmented generation (RAG) infrastructure designed for zero-trust enterprise environments. The pipeline ingests multi-format artifacts (PDF, DOCX, TXT) through an asynchronous parsing cluster, sanitizes sensitive content using Presidio PII anonymization routines, and partitions documentation using sentence-aware semantic chunking. Bounded contexts are assembled through reciprocal rank fusion (RRF) combining dense embeddings (1536-dimensional Gemini embedding-2 vectors) with sparse lexical BM25 token indices.

3. Agent Dispatch Protocols and Escalation Chains

The platform categorizes task execution into three operational tiers based on autonomous privilege:
- Tier-1 (Standard Read-Only): Permitted to query vector indices and generate summaries. Cannot execute system mutations.
- Tier-2 (Supervised Execution): Can propose schema modifications or document deletions, requiring asynchronous dual-approver confirmation.
- Tier-3 (Autonomous Core): Authorized to trigger autonomous query rerouting, knowledge graph entity re-indexing, and automated cache purges.

Escalation Protocol 409-B: In the event of a cluster partition or anomalous token drift exceeding 3.5%, the Lead Security Auditor (Elena Rostova) places the pipeline in Quarantine Mode. While in Quarantine Mode, the Agent Dispatcher is strictly forbidden from executing any Tier-3 autonomous actions EXCEPT under Condition 7-Alpha: if external API latency exceeds 4,500ms for more than three consecutive polling intervals AND the Chief Technology Officer digitally signs an emergency bypass token with SHA-384 hardware verification.

4. Technical Thresholds, Retrieval Benchmarks & SLA Matrix

The retrieval layer enforces strict deterministic latency budgets and mathematical thresholds across the vector database and semantic routing orchestrators. Deviations from these tolerances trigger automated telemetry alerts via Prometheus.

Metric / Parameter	Target Baseline	Critical Breach Threshold	Governing Protocol
Time To First Token (TTFT)	< 320 ms	> 850 ms	SLA-RT-01
Vector Cosine Threshold	0.78 similarity	< 0.62 similarity	RAG-GROUND-4
Reciprocal Rank Fusion k	k = 60 constant	k != 60	RRF-HYBRID-09
Presidio PII Anonymization	100% regex + NER	< 99.8% precision	COMPLIANCE-PII-2
Knowledge Graph Triple Ext.	Cosine > 0.82	Unlinked Triples > 5%	GRAPH-RAG-ENT-11
Cache TTL (Redis)	86,400 seconds (24h)	Cache Hit < 35%	CACHE-OPT-07

5. Routing Strategies & Edge-Case Abstention Criteria

All incoming queries are evaluated against four discrete semantic paths:
1. FACTUAL_VECTOR: Selected for single-document lookups where semantic density is above 0.78.
2. MULTI_HOP_GRAPH: Invoked whenever an inquiry spans cross-entity hierarchies, multi-tier dependencies, or organizational roles.
3. ANALYTICAL_SQL: Directed towards structured transactional logs, quantitative threshold comparisons, or numerical aggregations.
4. GREETING_DIRECT: Reserved for conversational pleasantries requiring zero document retrieval.

Abstention Requirement (Zero-Hallucination Policy): If retrieved passages contain conflicting data regarding operational permissions or if the highest reciprocal rank fusion score falls below 0.016, the system MUST NOT speculate. It is required to emit an explicit abstention response stating: "The provided documentation does not contain sufficient grounded evidence to substantiate this claim."

6. Disaster Recovery & Byzantine Partition Tolerance

Under Disaster Recovery Plan Omega (DRP-Omega), if primary cloud region US-East becomes unreachable, the vector store falls back to a read-only replica located in EU-Central within 1,200ms. During failover, all Tier-2 schema modification operations are queued in Redis with a maximum retention of 72 hours. Crucially, if the failover duration exceeds 180 minutes, the cryptographic audit trail requires physical re-keying by both Dr. Marcus Vance and Elena Rostova before full write privileges can be reinstated to the primary cluster.